



Advanced Cisco Networking Academy: Design and Configure a Multi-Area OSPF Network

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Purpose

The purpose of this lab is to be a refresher course to review the skills required to set up an OSPF network. Simultaneously, the lab adds a layer of complexity compared to the content taught in year 1 Cisco by making students learn how OSPF connects across multiple areas, and how to route IPv4/IPv6 traffic through a Layer 3 Switch.

Background

Open Shortest Path First is a routing protocol that uses link state advertisements to automatically build a network topology and provide end-to-end connectivity across networks. It was developed by the Internet Engineering Task Force, or IETF, a group that develops standards and operates under the Internet Society non-profit organization. This lab uses both OSPFv2 and OSPFv3, which operate similarly but use IPv4 and IPv6 addresses respectively. OSPFv2 is specified in RFC 2328 and OSPFv3 is specified in RFC 5320.

OSPF regularly updates its topology by detecting network changes such as dead routers. OSPF-enabled routers send hello messages to their neighbors at a given interval and assumes a neighboring router is dead after a longer interval. OSPF-enabled routers can have two different types of relationships with each other: neighbor relationships and adjacencies. Neighbor relationships, which are automatically formed in P2P, broadcast, and point-to-multipoint networks, are formed using hello messages and simply inform routers of each other's existence on the network. Adjacencies are a more complex type of relationship that allows routers to exchange routing information with each other.

The OSPF protocol has several different network types, each of which affect how relationships between routers are formed and maintained: point-to-point (P2P), broadcast, non-broadcast multi-access (NBMA), point-to-multipoint, and point-to-multipoint non-broadcast. A broadcast network is used by default and is the most common type of network. Broadcast networks and NBMA networks are the only type of networks to hold designated router elections. In a broadcast network, each router forms an adjacency to the designated and backup designated routers automatically. In an NBMA network, adjacencies to the DR/BDR can be formed, but neighbors must be manually configured before these adjacencies are formed. P2P and broadcast networks have hello timers of 10 seconds and dead timers of 40 seconds, while the other network types have hello timers of 30 seconds and dead timers of 120 seconds.

OSPF allows its network to be compartmentalized through areas. For the protocol to work, a backbone area (area 0 by default) must be set up and connected to all other areas. OSPF-enabled routers can have interfaces in more than one area as long as the next-hop router's connected interface is in the same area.

OSPF has several different types of routers which each work in harmony to connect different areas: backbone routers, internal routers, area border routers, and autonomous system border routers. OSPF routers can fall under one or more of these router types. Backbone routers have all their interfaces reside entirely within the backbone area. Internal routers have all their interfaces reside within one area that isn't the backbone. Area border routers have interfaces in more than one area and serve as an inter-area connection, maintaining separate routing information for each area. Autonomous system border routers maintain a connection to a network outside of the OSPF protocol. They can translate information from this outside network to OSPF routes.

Lab Summary

In this lab, we set up a multi-area OSPF network with three areas and seven total network devices. In area 0, we set up two area border routers and one area router. In area 1, we set up one border routers, with the internal router connected to a layer 3 multilayer switch. In area 2, we set up one internal router and one Layer 3 switch acting as an OSPF-enabled internal router, which also acted as the opposite network edge and was connected to another PC. We gave each network device a basic configuration with a hostname, passwords, and an MOTD, and assigned each a unique OSPF router ID. We then configured IPv4 and IPv6 addresses on each connected interface, opting to configure OSPF through the interfaces rather than using network statements.

Lab Commands

These are the commands to configure ospf through putty on all routers and switches. This is already assuming you have the base configuration (Passwords, hostnames, and banners) done on each router and switch. These commands are the commands for router 3 in this lab.

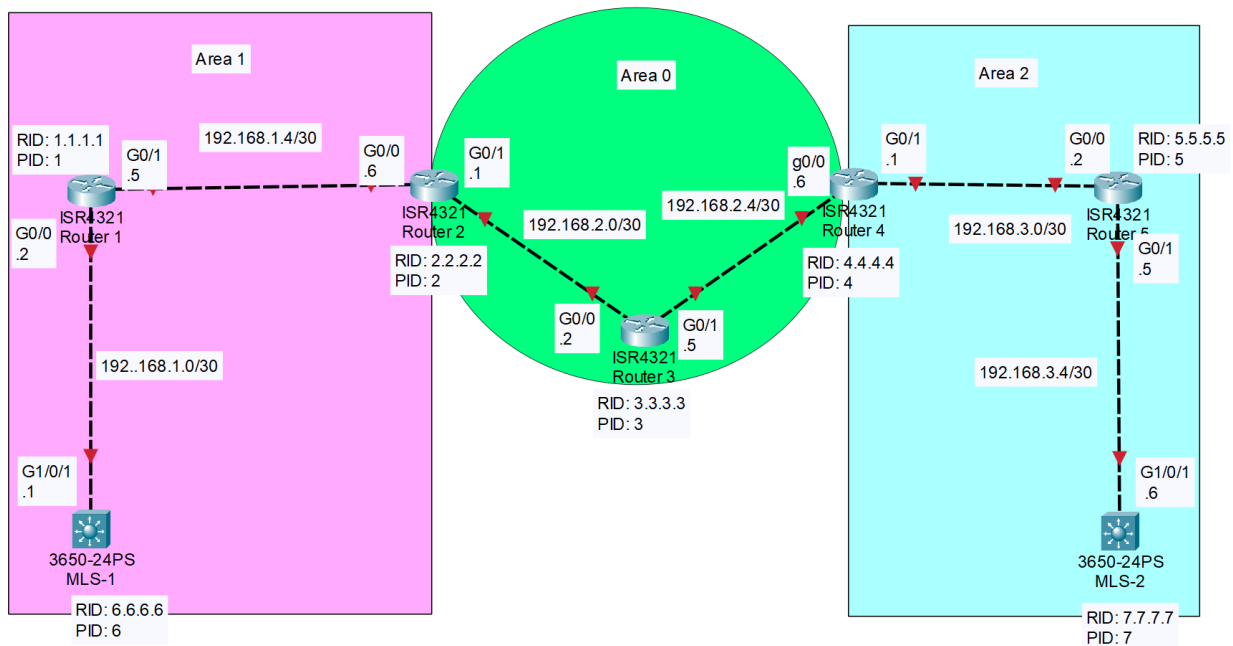
1. Enables IP services on a switch for IPv4 and IPv6. These commands must be entered on a the config mode on the router/switch before IP addresses can be configured.
`Router3(config)#ipv6 unicast-routing`
`Router3(config)#ip routing`
2. Configure OSPF for IPV4 with a process ID of 1, and configures a router ID
`Router3(config)#router ospf 1`
`Router3(config-router)#router-id <id>`
3. Configures OSPF for IPV6 with a process ID of 1, and configures a router ID.
`Switch(config)# ipv6 router ospf 1`
`Switch(config-rtr)# router-id <id>`
4. Configure iIPv4 and iIPv6 address in each interface on all routers and switches
`Router3(config-if)#ip address 192.168.2.2 255.255.255.252`
`Router3(config-if)#ipv6 address 2001:db8:67:abcd::2/36`

5. Configures OSPF on an interface for both IPv4 and IPv6 with a process ID of 1 and a specified area ID.

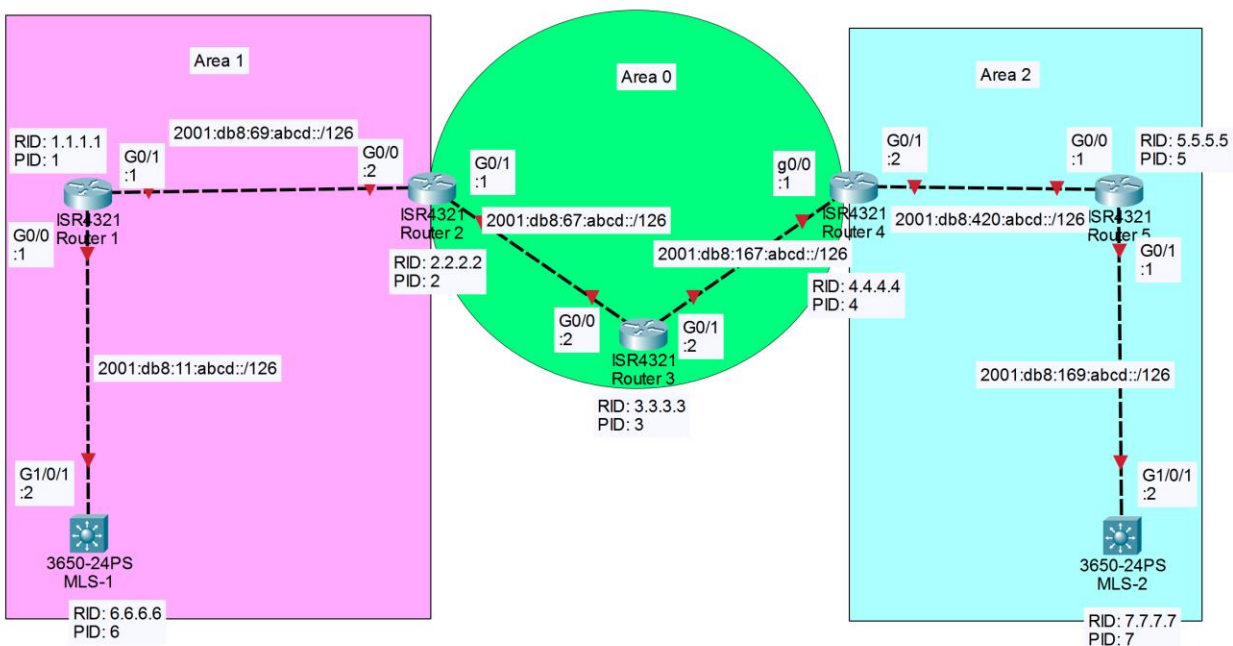
```
Router3(config-if)#ip ospf 1 area 0
Router3(config-if)#ipv6 ospf 1 area 0
```

Network Diagrams

IPv4 Network Topology



IPv6 Network Topology



Lab Configurations

R1:

```
Current configuration : 1661 bytes
!
! Last configuration change at 18:15:28 UTC Fri Sep 12 2025
!
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
!
hostname Router1
!
boot-start-marker
boot-end-marker
!
!
vrf definition Mgmt-intf
!
  address-family ipv4
  exit-address-family
!
  address-family ipv6
  exit-address-family
!
!
no aaa new-model
!
!
login on-success log
!
!
subscriber templating
ipv6 unicast-routing
multilink bundle-name authenticated
!
!
license udi pid ISR4321/K9 sn FLM2408005M
no license smart enable
diagnostic bootup level minimal
```

```

!
spanning-tree extend system-id
!
!
redundancy
mode none
!
!
interface GigabitEthernet0/0/0
ip address 192.168.1.2 255.255.255.252
ip ospf 1 area 1
negotiation auto
ipv6 address 2001:DB8:11:ABCD::1/126
ipv6 ospf 1 area 1
!
interface GigabitEthernet0/0/1
ip address 192.168.1.5 255.255.255.252
ip ospf 1 area 1
negotiation auto
ipv6 address 2001:DB8:69:ABCD::1/126
ipv6 ospf 1 area 1
!
interface GigabitEthernet0/2/0
no ip address
negotiation auto
!
interface GigabitEthernet0/2/1
no ip address
negotiation auto
!
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
negotiation auto
!
router ospf 1
router-id 1.1.1.1
!
ip forward-protocol nd
no ip http server
ip http secure-server
ip tftp source-interface GigabitEthernet0
!
!
ipv6 router ospf 1
router-id 1.1.1.1
!

```



```
!  
!  
!  
control-plane  
!  
banner motd ^Cdo not enter^C  
!  
line con 0  
  password cisco  
  login  
  transport input none  
  stopbits 1  
line aux 0  
  stopbits 1  
line vty 0 4  
  login  
!  
!  
End
```

R2:

```
Current configuration : 1661 bytes  
!  
! Last configuration change at 17:42:35 UTC Fri Sep 12 2025  
!  
version 16.9  
service timestamps debug datetime msec  
service timestamps log datetime msec  
platform qfp utilization monitor load 80  
platform punt-keepalive disable-kernel-core  
!  
hostname Router2  
!  
boot-start-marker  
boot-end-marker  
!  
!  
vrf definition Mgmt-intf  
!  
  address-family ipv4  
  exit-address-family  
!  
  address-family ipv6  
  exit-address-family  
!  
!  
no aaa new-model
```

```

!
!
!
!
login on-success log
!
!
subscriber templating
ipv6 unicast-routing
multilink bundle-name authenticated
!
!
license udi pid ISR4321/K9 sn FLM24060912
no license smart enable
diagnostic bootup level minimal
!
spanning-tree extend system-id
!
!
redundancy
mode none
!
!
interface GigabitEthernet0/0/0
ip address 192.168.1.6 255.255.255.252
ip ospf 1 area 1
negotiation auto
ipv6 address 2001:DB8:69:ABCD::2/126
ipv6 ospf 1 area 1
!
interface GigabitEthernet0/0/1
ip address 192.168.2.1 255.255.255.252
ip ospf 1 area 0
negotiation auto
ipv6 address 2001:DB8:67:ABCD::1/126
ipv6 ospf 1 area 0
!
interface GigabitEthernet0/2/0
no ip address
negotiation auto
!
interface GigabitEthernet0/2/1
no ip address
negotiation auto
!
interface GigabitEthernet0
vrf forwarding Mgmt-intf

```



```

no ip address
negotiation auto
!
router ospf 1
router-id 2.2.2.2
!
ip forward-protocol nd
no ip http server
ip http secure-server
ip tftp source-interface GigabitEthernet0
!
!
ipv6 router ospf 1
router-id 2.2.2.2
!
!
control-plane
!
banner motd ^Cdo not enter^C
!
line con 0
password cisco
login
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
!
!
end

```

R3:

```

Current configuration : 1618 bytes
!
! Last configuration change at 18:33:50 UTC Thu Sep 11 2025
!
version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
no platform punt-keepalive disable-kernel-core
!
hostname Router3
!
boot-start-marker

```

```
boot-end-marker
!
!
vrf definition Mgmt-intf
!
  address-family ipv4
  exit-address-family
!
  address-family ipv6
  exit-address-family
!
!
no aaa new-model
!
ipv6 unicast-routing
!
!
subscriber templating
multilink bundle-name authenticated
!
!
!
!
license udi pid ISR4321/K9 sn FDO214420G7
!
spanning-tree extend system-id
!
!
redundancy
  mode none
!
!
vlan internal allocation policy ascending
!
!
interface GigabitEthernet0/0/0
  ip address 192.168.2.2 255.255.255.252
  ip ospf 1 area 0
  negotiation auto
  ipv6 address 2001:DB8:67:ABCD::2/126
  ipv6 ospf 1 area 0
!
interface GigabitEthernet0/0/1
  ip address 192.168.2.5 255.255.255.252
  ip ospf 1 area 0
  negotiation auto
  ipv6 address 2001:DB8:167:ABCD::2/126
```

```

    ipv6 ospf 1 area 0
!
interface Serial0/1/0
    no ip address
    shutdown
!
interface Serial0/1/1
    no ip address
    shutdown
!
interface GigabitEthernet0
    vrf forwarding Mgmt-intf
    no ip address
    shutdown
    negotiation auto
!
interface Vlan1
    no ip address
    shutdown
!
router ospf 1
    router-id 3.3.3.3
!
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
!
!
ipv6 router ospf 1
    router-id 3.3.3.3
!
!
!
!
control-plane
!
!
line con 0
    stopbits 1
line aux 0
    stopbits 1
line vty 0 4
    login
!
!
End

```

R4

```
Current configuration : 1609 bytes
!
! Last configuration change at 17:27:02 UTC Fri Sep 12 2025
!
version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
no platform punt-keepalive disable-kernel-core
!
hostname R4
!
boot-start-marker
boot-end-marker
!
!
vrf definition Mgmt-intf
!
    address-family ipv4
        exit-address-family
    !
    address-family ipv6
        exit-address-family
!
!
no aaa new-model
!
!

!
!
!
!
ipv6 unicast-routing
!
!
subscriber templating
multilink bundle-name authenticated
!
!
!
!
license udi pid ISR4321/K9 sn FDO211216BL
!
```

```
spanning-tree extend system-id
!
!
redundancy
mode none
!
!
vlan internal allocation policy ascending
!
!
interface GigabitEthernet0/0/0
ip address 192.168.3.1 255.255.255.252
ip ospf 1 area 0
negotiation auto
ipv6 address 2001:DB8:420:ABCD::2/126
ipv6 ospf 1 area 0
!
interface GigabitEthernet0/0/1
ip address 192.168.2.6 255.255.255.252
negotiation auto
ipv6 address 2001:DB8:167:ABCD::1/126
ipv6 ospf 1 area 2
!
interface Serial0/1/0
no ip address
!
interface Serial0/1/1
no ip address
!
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
negotiation auto
!
interface Vlan1
no ip address
!
router ospf 1
router-id 4.4.4.4
!
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
!
!
ipv6 router ospf 1
```

```
router-id 4.4.4.4
!  
!  
control-plane
!  
!  
line con 0
password 7 1511021F0725
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
!  
!  
end
```

R5

```
Current configuration : 1516 bytes
!  
version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
no platform punt-keepalive disable-kernel-core
!  
hostname Router
!  
boot-start-marker
boot-end-marker
!  
!  
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-famil
!  
!  
no aaa new-model
!  
!  
ipv6 unicast-routing
!  
!  
!  
!  
!
```

```

!
!
subscriber templating
multilink bundle-name authenticated
!
!
license udi pid ISR4321/K9 sn FDO214421CF
!
spanning-tree extend system-id
!
!
redundancy
mode none
!
!
vlan internal allocation policy ascendin
!
!
interface GigabitEthernet0/0/0
ip address 192.168.3.2 255.255.255.252
ip ospf 1 area 2
negotiation auto
ipv6 address 2001:DB8:420:ABCD::1/126
ipv6 ospf 1 area 2
!
interface GigabitEthernet0/0/1ip address 192.168.3.5
255.255.255.252
ip ospf 1 area 2
negotiation auto
ipv6 address 2001:DB8:169:ABCD::1/126
ipv6 ospf 1 area 2
!
interface Serial0/1/0
no ip address
!
interface Serial0/1/
no ip address
!
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip adres
negotiation auto
!
interface Vlan1
no ip address
!
router ospf 1

```



```
router-id 5.5.5.5
!
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
!
!
ipv6 router ospf 1
router-id 5.5.5.5
!
!
control-plane
!
!
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
!
!
End
```

MLS1

```
Current configuration : 9213 bytes
!
! Last configuration change at 17:38:44 UTC Fri Sep 12 2025
!
version 16.12
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
! Call-home is enabled by Smart-Licensing.
service call-home
platform punt-keepalive disable-kernel-core
!
hostname switch1
!
!
vrf definition Mgmt-vrf
!
address-family ipv4
exit-address-family
!
address-family ipv6
```

```

    exit-address-family
!
!
!
!
!
no aaa new-model
switch 1 provision c92001-24t-4g
!
!
ip routing
!
!
!
!
login on-success log
ipv6 unicast-routing
call-home
    ! If contact email address in call-home is configured as sch-
    smart-licensing@cisco.com
    ! the email address configured in Cisco Smart License Portal
    will be used as contact email address to send SCH notifications.
    contact-email-addr sch-smart-licensing@cisco.com
    profile "CiscoTAC-1"
        active
        destination transport-method http
        no destination transport-method email
no device-tracking logging theft
!
diagnostic bootup level minimal
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
memory free low-watermark processor 10055
!
!
redundancy
    mode sso
!
!
interface GigabitEthernet0/0
    vrf forwarding Mgmt-vrf
    no ip address
    shutdown
    negotiation auto
!
interface GigabitEthernet1/0/1

```

```
no switchport
ip address 192.168.1.1 255.255.255.252
ip ospf 1 area 1
ipv6 address 2001:DB8:11:ABCD::2/126
ipv6 ospf 1 area 1
!
interface GigabitEthernet1/0/2
!
interface GigabitEthernet1/0/3
!
interface GigabitEthernet1/0/4
!
interface GigabitEthernet1/0/5
!
interface GigabitEthernet1/0/6
!
interface GigabitEthernet1/0/7
!
interface GigabitEthernet1/0/8
!
interface GigabitEthernet1/0/9
!
interface GigabitEthernet1/0/10
!
interface GigabitEthernet1/0/11
!
interface GigabitEthernet1/0/12
!
interface GigabitEthernet1/0/13
!
interface GigabitEthernet1/0/14
!
interface GigabitEthernet1/0/15
!
interface GigabitEthernet1/0/16
!
interface GigabitEthernet1/0/17
!
interface GigabitEthernet1/0/18
!
interface GigabitEthernet1/0/19
!
interface GigabitEthernet1/0/20
!
interface GigabitEthernet1/0/21
!
interface GigabitEthernet1/0/22
```

```

!
interface GigabitEthernet1/0/23
!
interface GigabitEthernet1/0/24
!
interface GigabitEthernet1/1/1
!
interface GigabitEthernet1/1/2
!
interface GigabitEthernet1/1/3
!
interface GigabitEthernet1/1/4
!
interface Vlan1
  no ip address
!
router ospf 1
  router-id 6.6.6.6
!
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
!
!
ipv6 router ospf 1
  router-id 6.6.6.6
!
!
!
control-plane
  service-policy input system-cpp-policy
!
!
line con 0
  password cisco
  login
  stopbits 1
line aux 0
  stopbits 1
line vty 0 4
  login
line vty 5 15
  login
!
!
!

```

```
!  
!  
!  
!  
end
```

MLS2

```
Current configuration : 8657 bytes  
!  
! Last configuration change at 18:06:46 UTC Fri Sep 12 2025  
!  
version 16.9  
no service pad  
service timestamps debug datetime msec  
service timestamps log datetime msec  
! Call-home is enabled by Smart-Licensing.  
service call-home  
no platform punt-keepalive disable-kernel-core  
!  
hostname s2  
!  
!  
vrf definition Mgmt-vrf  
!  
  address-family ipv4  
  exit-address-family  
!  
  address-family ipv6  
  exit-address-family  
!  
!  
!  
!  
!  
no aaa new-model  
switch 1 provision c9200l-24t-4g  
!  
!  
!  
!  
!  
!  
!  
!  
!
```

```

ip routing
ip arp entry learn 10240
!
!
!
!
login on-success log
ipv6 unicast-routing
call-home
  ! If contact email address in call-home is configured as sch-
  smart-licensing@cisco.com
  ! the email address configured in Cisco Smart License Portal
  will be used as contact email address to send SCH notifications.
  contact-email-addr sch-smart-licensing@cisco.com
  profile "CiscoTAC-1"
    active
    destination transport-method http
    no destination transport-method email
  !
crypto pki trustpoint SLA-TrustPoint
  enrollment pkcs12
  revocation-check crl
!
crypto pki trustpoint TP-self-signed-2221393201
  enrollment selfsigned
  subject-name cn=IOS-Self-Signed-Certificate-2221393201
  revocation-check none
  rsakeypair TP-self-signed-2221393201
!
license boot level network-essentials addon dna-essentials
!
!
diagnostic bootup level minimal
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
!
!
redundancy
  mode sso
!
!
policy-map system-cpp-policy
!
!
!

```

```
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
interface GigabitEthernet0/0  
  vrf forwarding Mgmt-vrf  
  no ip address  
  shutdown  
  negotiation auto  
!  
interface GigabitEthernet1/0/1  
  no switchport  
  ip address 192.168.3.6 255.255.255.252  
  ip ospf 1 area 2  
  ipv6 address 2001:DB8:169:ABCD::2/126  
  ipv6 ospf 1 area 2  
!  
interface GigabitEthernet1/0/2  
!  
interface GigabitEthernet1/0/3  
!  
interface GigabitEthernet1/0/4  
!  
interface GigabitEthernet1/0/5  
!  
interface GigabitEthernet1/0/6  
!  
interface GigabitEthernet1/0/7  
!  
interface GigabitEthernet1/0/8  
!  
interface GigabitEthernet1/0/9  
!  
interface GigabitEthernet1/0/10  
!  
interface GigabitEthernet1/0/11  
!  
interface GigabitEthernet1/0/12  
!  
interface GigabitEthernet1/0/13  
!
```

```
interface GigabitEthernet1/0/14
!
interface GigabitEthernet1/0/15
!
interface GigabitEthernet1/0/16
!
interface GigabitEthernet1/0/17
!
interface GigabitEthernet1/0/18
!
interface GigabitEthernet1/0/19
!
interface GigabitEthernet1/0/20
!
interface GigabitEthernet1/0/21
!
interface GigabitEthernet1/0/22
!
interface GigabitEthernet1/0/23
!
interface GigabitEthernet1/0/24
!
interface GigabitEthernet1/1/1
!
interface GigabitEthernet1/1/2
!
interface GigabitEthernet1/1/3
!
interface GigabitEthernet1/1/4
!
interface Vlan1
  no ip address
  shutdown
!
router ospf 1
  router-id 7.7.7.7
!
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
!
!
ipv6 router ospf 1
  router-id 7.7.7.7
!
!
```



```

!
control-plane
  service-policy input system-cpp-policy
!
!
line con 0
  stopbits 1
line aux 0
  stopbits 1
line vty 0 4
  login
line vty 5 15
  login
!
!
!
end

```

Testing:

IPv4: Traceroute from MLS1-MLS2: traceroute 192.168.3.6

Tracing the route to 192.168.3.6

VRF info: (vrf in name/id, vrf out name/id)

```

 1 192.168.1.2 4 msec 4 msec 0 msec
 2 192.168.1.6 0 msec 4 msec 0 msec
 3 192.168.2.2 0 msec 4 msec 0 msec
 4 192.168.2.6 0 msec 4 msec 0 msec
 5 192.168.3.2 4 msec 0 msec 4 msec
 6 192.168.3.6 112 msec 4 msec *
```

IPv4: Traceroute from MLS2-MLS1: traceroute 192.168.1.1

Tracing the route to 192.168.1.1

VRF info: (vrf in name/id, vrf out name/id)

```

 1 192.168.3.5 20 msec 4 msec 0 msec
 2 192.168.3.1 0 msec 4 msec 0 msec
 3 192.168.2.5 0 msec 4 msec 0 msec
 4 192.168.2.1 4 msec 4 msec 0 msec
 5 192.168.1.5 0 msec 0 msec 4 msec
 6 192.168.1.1 0 msec 4 msec *
```

Ipv6: Tracerout from ML2-MLS1: traceroute 2001:db8:11:abcd::2

Tracing the route to 2001:DB8:11:ABCD::2

```

 1 2001:DB8:169:ABCD::1 0 msec 0 msec 0 msec
 2 2001:DB8:420:ABCD::2 4 msec 0 msec 0 msec
 3 2001:DB8:167:ABCD::2 4 msec 0 msec 0 msec
 4 2001:DB8:67:ABCD::1 4 msec 0 msec 4 msec
```

```

5 2001:DB8:69:ABCD::1 0 msec 4 msec 0 msec
6 2001:DB8:11:ABCD::2 24 msec 4 msec 0 msec

```

Ipv6: Tracerout from ML1-MLS2: traceroute 2001:db8:169:abcd::2

Tracing the route to 2001:DB8:169:ABCD::2

```

1 2001:DB8:11:ABCD::1 4 msec 0 msec 0 msec
2 2001:DB8:69:ABCD::2 4 msec 4 msec 0 msec
3 2001:DB8:67:ABCD::2 4 msec 4 msec 0 msec
4 2001:DB8:167:ABCD::1 0 msec 4 msec 0 msec
5 2001:DB8:420:ABCD::1 4 msec 4 msec 0 msec
6 2001:DB8:169:ABCD::2 0 msec 4 msec 0 msec

```

Show Ip OSPF 1 Database on switch 2:

It shows multiple area numbers.

Show Ip OSPF 1 Database on switch 1:

OSPF Router with ID (6.6.6.6) (Process ID 1)

Router Link States (Area 1)

Link ID	ADV Router	Age	Seq#	Checksum
Link count				
1.1.1.1	1.1.1.1	755	0x80000005	0x007EE0
2				
2.2.2.2	2.2.2.2	101	0x80000004	0x00A797
1				
6.6.6.6	6.6.6.6	745	0x80000003	0x001118
1				

Net Link States (Area 1)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.1.2	1.1.1.1	756	0x80000001	0x00EDC2
192.168.1.5	1.1.1.1	816	0x80000001	0x0007B6

Summary Net Link States (Area 1)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.2.0	2.2.2.2	101	0x80000001	0x007D4E
192.168.2.4	2.2.2.2	57	0x80000001	0x005F67
192.168.3.0	2.2.2.2	51	0x80000001	0x008642
192.168.3.4	2.2.2.2	51	0x80000001	0x00685B

show Ip OSPF 1 Database on router 1:

OSPFv3 1 address-family ipv6 (router-id 192.168.1.5)

Router Link States (Area 1)

ADV Router	Age	Seq#	Fragment ID	Link count	Bits
11.11.11.11	3076	0x80000013	0	1	None
192.168.1.1	1018	0x80000004	0	1	None
192.168.1.5	1114	0x8000000A	0	2	None
192.168.2.1	533	0x80000005	0	1	B

Net Link States (Area 1)

ADV Router	Age	Seq#	Link ID	Rtr count
192.168.1.5	1114	0x80000004	6	2
192.168.1.5	1114	0x80000002	7	2

Inter Area Prefix Link States (Area 1)

ADV Router	Age	Seq#	Prefix
192.168.2.1	533	0x80000002	2001:DB8:67:ABCD::/126
192.168.2.1	277	0x80000002	2001:DB8:167:ABCD::/126

Link (Type-8) Link States (Area 1)

ADV Router	Age	Seq#	Link ID	Interface
192.168.1.5	1114	0x80000003	7	Gi0/0/1
192.168.2.1	1051	0x80000003	6	Gi0/0/1
11.11.11.11	3155	0x80000002	38	Gi0/0/0
192.168.1.1	1018	0x80000003	38	Gi0/0/0
192.168.1.5	1114	0x80000003	6	Gi0/0/0

Intra Area Prefix Link States (Area 1)

ADV Router LSID	Age	Seq#	Link ID	Ref-lsttype	Ref-
192.168.1.5	1114	0x80000004	6144	0x2002	6
192.168.1.5	1114	0x80000002	7168	0x2002	7

show Ip OSPF 1 Database on router 2:

OSPF Router with ID (2.2.2.2) (Process ID 1)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
Link count				
2.2.2.2	2.2.2.2	711	0x80000005	0x0057EC
1				

```

3.3.3.3      3.3.3.3      810      0x80000009 0x002C1A
2
4.4.4.4      4.4.4.4      732      0x80000004 0x002FFC
1

```

Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.2.2	3.3.3.3	810	0x80000002	0x00208E
192.168.2.6	4.4.4.4	732	0x80000002	0x002E70

Summary Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.1.0	2.2.2.2	711	0x80000002	0x00903A
192.168.1.4	2.2.2.2	711	0x80000002	0x005E69
192.168.3.0	4.4.4.4	732	0x80000002	0x00348D
192.168.3.4	4.4.4.4	732	0x80000002	0x0016A6

Router Link States (Area 1)

Link ID	ADV Router	Age	Seq#	Checksum
Link count				
1.1.1.1	1.1.1.1	1450	0x80000006	0x007CE1
2				
2.2.2.2	2.2.2.2	711	0x80000005	0x00A598
1				
6.6.6.6	6.6.6.6	1357	0x80000004	0x000F19
1				

Net Link States (Area 1)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.1.2	1.1.1.1	1450	0x80000002	0x00EBC3
192.168.1.5	1.1.1.1	1450	0x80000002	0x0005B7

Summary Net Link States (Area 1)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.2.0	2.2.2.2	711	0x80000002	0x007B4F
192.168.2.4	2.2.2.2	711	0x80000002	0x005D68
192.168.3.0	2.2.2.2	711	0x80000002	0x008443
192.168.3.4	2.2.2.2	711	0x80000002	0x00665C

show Ip OSPF 1 Database on router 3:

OSPF Router with ID (3.3.3.3) (Process ID 1)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
Link count				
2.2.2.2	2.2.2.2	748	0x80000005	0x0057EC
1				
3.3.3.3	3.3.3.3	845	0x80000009	0x002C1A
2				
4.4.4.4	4.4.4.4	767	0x80000004	0x002FFC
1				

Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.2.2	3.3.3.3	845	0x80000002	0x00208E
192.168.2.6	4.4.4.4	767	0x80000002	0x002E70

Summary Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.1.0	2.2.2.2	748	0x80000002	0x00903A
192.168.1.4	2.2.2.2	748	0x80000002	0x005E69
192.168.3.0	4.4.4.4	767	0x80000002	0x00348D
192.168.3.4	4.4.4.4	767	0x80000002	0x0016A6

show Ip OSPF 1 Database on router 4:

OSPF Router with ID (4.4.4.4) (Process ID 1)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
Link count				
2.2.2.2	2.2.2.2	371	0x80000014	0x0039FB
1				
3.3.3.3	3.3.3.3	52	0x80000009	0x002C1A
2				
4.4.4.4	4.4.4.4	49	0x80000004	0x002FFC
1				

Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.2.2	3.3.3.3	378	0x80000002	0x00208E
192.168.2.6	4.4.4.4	54	0x80000001	0x00306F

Summary Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.1.0	2.2.2.2	419	0x80000001	0x009239
192.168.1.4	2.2.2.2	419	0x80000001	0x006068
192.168.3.0	4.4.4.4	99	0x80000001	0x00368C
192.168.3.4	4.4.4.4	99	0x80000001	0x0018A5

Router Link States (Area 2)

Link ID	ADV Router	Age	Seq#	Checksum
Link count				
4.4.4.4	4.4.4.4	99	0x80000007	0x00D05F
1				
5.5.5.5	5.5.5.5	1081	0x8000000A	0x00D55B
2				
7.7.7.7	7.7.7.7	1434	0x80000009	0x0053BA
1				

Net Link States (Area 2)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.3.2	5.5.5.5	1096	0x80000001	0x008313
192.168.3.6	7.7.7.7	1438	0x80000001	0x0095E8

Summary Net Link States (Area 2)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.1.0	4.4.4.4	54	0x80000001	0x006A57
192.168.1.4	4.4.4.4	54	0x80000001	0x003886
192.168.2.0	4.4.4.4	54	0x80000001	0x004B77
192.168.2.4	4.4.4.4	99	0x80000001	0x0019A6

show Ip OSPF 1 Database on router 5:

OSPF Router with ID (5.5.5.5) (Process ID 1)

Router Link States (Area 2)

Link ID	ADV Router	Age	Seq#	Checksum
Link count				
4.4.4.4	4.4.4.4	810	0x80000008	0x00CE60
1				
5.5.5.5	5.5.5.5	967	0x80000011	0x00C762
2				
7.7.7.7	7.7.7.7	867	0x80000006	0x0059B7
1				

Net Link States (Area 2)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.3.2	5.5.5.5	1475	0x80000002	0x008114
192.168.3.6	7.7.7.7	867	0x80000002	0x0093E9

Summary Net Link States (Area 2)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.1.0	4.4.4.4	810	0x80000002	0x006858
192.168.1.4	4.4.4.4	810	0x80000002	0x003687
192.168.2.0	4.4.4.4	810	0x80000002	0x004978
192.168.2.4	4.4.4.4	810	0x80000002	0x0017A7

Show Ip route on MLS 2; The IA indicates that the routing happens Inter-area, which shows the evidence of multi-area ospf.

```
192.168.1.0/30 is subnetted, 2 subnets
O IA      192.168.1.0 [110/6] via 192.168.3.5, 00:21:27,
GigabitEthernet1/0/1
O IA      192.168.1.4 [110/5] via 192.168.3.5, 00:21:27,
GigabitEthernet1/0/1
    192.168.2.0/30 is subnetted, 2 subnets
O IA      192.168.2.0 [110/4] via 192.168.3.5, 00:21:27,
GigabitEthernet1/0/1
O IA      192.168.2.4 [110/3] via 192.168.3.5, 00:22:07,
GigabitEthernet1/0/1
    192.168.3.0/24 is variably subnetted, 3 subnets, 2 masks
O        192.168.3.0/30
        [110/2] via 192.168.3.5, 00:23:30,
GigabitEthernet1/0/1
C        192.168.3.4/30 is directly connected,
GigabitEthernet1/0/1
L        192.168.3.6/32 is directly connected,
GigabitEthernet1/0/1
```

Show ip route on MLS 1:

```
192.168.1.0/24 is variably subnetted, 3 subnets, 2 masks
C        192.168.1.0/30 is directly connected,
GigabitEthernet1/0/1
L        192.168.1.1/32 is directly connected,
GigabitEthernet1/0/1
O        192.168.1.4/30
```

```
        [110/2] via 192.168.1.2, 00:37:04,
GigabitEthernet1/0/1
        192.168.2.0/30 is subnetted, 2 subnets
O IA      192.168.2.0 [110/3] via 192.168.1.2, 00:26:09,
GigabitEthernet1/0/1
O IA      192.168.2.4 [110/4] via 192.168.1.2, 00:25:25,
GigabitEthernet1/0/1
        192.168.3.0/30 is subnetted, 2 subnets
O IA      192.168.3.0 [110/5] via 192.168.1.2, 00:25:19,
GigabitEthernet1/0/1
O IA      192.168.3.4 [110/6] via 192.168.1.2, 00:25:19,
GigabitEthernet1/0/1
```

Problems

The versions for the routers we are using are different. The software is different, so some interfaces will not be able to ping some interfaces. To mitigate this, we switched the routers to routers with the correct version on them, so all the routers in the network are the same version at 16.9.

For one of the routers, the IPv6 ospf was not configured on one of the interfaces. We found this issue in the testing phase, when we were looking at OSPF databases for all the routers. We noticed that in the OSPF 1PV6 database, the router id for 2 did not show up causing connection issues. To fix this, we simply went into the interfaces and configured IPv6 OSPF on the interfaces.

Conclusion